

Enhanced incentive schemes for volunteers to distribute surplus food in smart cities

Surja Sanyal ^{a,*}, Sajal Mukhopadhyay ^b, Fatos Xhafa ^c

^a Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, Guwahati, 781039, Assam, India

^b Department of Computer Science and Engineering, National Institute of Technology Durgapur, Durgapur, 713209, West Bengal, India

^c Department of Computer Science, Universitat Politècnica de Catalunya, Barcelona, 08034, Catalonia, Spain

ARTICLE INFO

Dataset link: <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us>

Keywords:

Food wastage
Food recovery
Surplus food distribution
Volunteer incentive
Volunteer participation
Sybil attack

ABSTRACT

Food wastage is a major problem in today's world and the distribution of food surplus is undoubtedly one of the leading solutions to the problem. Surplus distribution is widely practised across the globe and there are several communities for the same as well. However, the success of these movements are determined by the active participation of volunteers. In addition, volunteer availability is an issue that is dependent on several different factors. An incentive module for the volunteers can normalize the availability of the volunteers. Such incentive modules are prone to identity duplication attacks and the implemented solutions do not really encourage volunteer hierarchical structures. In this paper, an incentive module that resists identity duplication and promotes the hierarchical structure between volunteers is presented. In addition, an improved matching module to match the food donors to the receivers utilizing the available volunteers optimally is also proposed. For the simulation, we have used real data taken from a real life smart city context, namely, we have used the OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us> and cast it to our simulation settings using Python libraries.

1. Introduction

The global issue of food wastage has been dealt with by the help of surplus distribution. This approach has been accepted as the most viable solution to the challenge. The usage of technological platforms available in existing and upcoming smart cities have rendered this approach highly scalable and naturally sustainable in the long run. However, the success of such movements relies heavily on the participation of volunteers. That, in turn, is dependent on various factors including, but not limited to, the season, the time of the day, the weather conditions, ongoing festivities, etc. The Food Donor to Receiver Matching with Chronological Acceptance (FDRM-CA) was proposed in our work [Sanyal, Kumar Singh, Xhafa, Sanyal, and Mukhopadhyay \(2021\)](#). An incentive module was developed to encourage volunteer participation as well ([Sanyal, Mukhopadhyay, Xhafa, Kumar Singh, & Sanyal, 2022](#)).

However, it was noted that the presented incentive scheme was susceptible to identity duplication attacks and the solution implemented therein did not particularly favour hierarchical volunteer engagement in distribution activities. Here, in this work, we present an incentive module that resists such identity duplication attacks and encourages volunteer hierarchical work structure. Furthermore, it also rewards

volunteers for promoting the scheme to other volunteers. We also improve upon the matching module so that it can produce better results under tighter volunteer availability conditions. For the simulation, we have used real data taken from a real life smart city context, namely, we have used the OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us> and cast it to our simulation settings using Python libraries.

The major contributions of this paper start with the modified food donor to receiver matching that includes a larger pool of volunteers for each surplus donor from where an optimal volunteer is assigned. This increases volunteer utilization leading to a higher surplus mobility. In addition, an incentive mechanism is proposed for the volunteers to attract better participation in moving the surplus food. Two methods for the disbursement of volunteer incentives are suggested that are suitable for different kinds of volunteer incentive setup requirements. One of the methods favours the volunteers in terms of a higher incentive amount. However, the volunteers may gain more incentive by using duplicate identities. The other method favours the system in terms of being resistant to identity duplication at the cost of lower incentives disbursed to the volunteers. Either method stabilizes the availability of volunteers over the different factors (e.g., the season, the time of the day, etc.)

* Corresponding author.

E-mail addresses: hi.surja06@gmail.com (S. Sanyal), sajal@cse.nitdgp.ac.in (S. Mukhopadhyay), fatos@cs.upc.edu (F. Xhafa).

<https://doi.org/10.1016/j.scs.2023.104409>

Received 13 July 2022; Received in revised form 22 December 2022; Accepted 16 January 2023

Available online 20 January 2023

2210-6707/© 2023 Elsevier Ltd. All rights reserved.

resulting in the robust performance of the proposed mechanism. The modified matching mechanism along with the two proposed incentive disbursal methods produce superior results when compared to that of the existing method.

The implementation of the mechanisms put forward in this paper will enable the smart cities to have a much more resilient environment as the downtrodden people of the society can lead a better life by acquiring the food they require on a daily basis. Supply of food to the needy one will have a feel-good impact socially as a large section of the society will be involved in this process in terms of donors, volunteers, and receivers. As the eatable food is being consumed instead of being thrown away, it can be said that waste is being reduced implicitly.

This paper is organized as follows. Section 2 presents the related work in this field. The system model and the proposed mechanism have been discussed in Section 3. The mechanism has been analyzed in Section 4 and its simulation results presented in Section 5. A discussion about the outcomes of the simulations has been put forward in Section 6 followed by conclusion and future work in Section 7.

2. Related work

We have already provided a food donor to receiver matching algorithm (Sanyal et al., 2021) and a volunteer incentive generation algorithm (Sanyal et al., 2022). This paper aims at improving the incentive mechanism so as to make it better immune to identity duplication attacks (sybil attacks). It also modifies the matching mechanism so that it produces optimal matches under reduced volunteer availability conditions.

Wastage of food is the top concern for people across the globe as it leads to several other social and economical issues (Ganglbauer, Fitzpatrick, & Molzer, 2012). Amounting close to one third of the food production, most of this billions of tonnes of wasted food is still in perfectly edible condition when being discarded (Shi, Yuan, Lo, Lizarondo, & Fang, 2020; Zhong, Xu, & Wang, 2017). Solutions to this issue are several. However, the distribution of surplus food works for this issue as well as the issue of food supply to those in need.

Far from new, the distribution of surplus food is already observable in the society with credible support from the Information and Communications Technology (ICT) platforms (Esteso, Alemany, & Ortiz, 2021). Further useful contributions are noted from the usage of the Internet of Things (IoT) in the terms of traceability and accountability (Nagarajan, Deverajan, Chatterjee, Alnumay, & Muthukumaran, 2022).

The steep growth of ICT based platforms is well documented and the impact of food donation on the society is also captured (Spring & Biddulph, 2020). The reverse of the same, that is, the impact of society and culture on the donation of food, is also explored (Rut, Davies, & Ng, 2020). Food distribution has also been pointed out as a solution to the rampant challenge of global climate change (Nikravech, Kwan, Dobernig, Wilhelm-Rechmann, & Langen, 2020). Volunteer participation and ICT platform usage are the top reasons for the success of such low running-cost ventures. A centralized food collection and distribution body goes a long way in making the food distribution activity more sustainable (Bergström, Malefors, Strid, Hanssen, & Eriksson, 2020). The economically better approach of food distributed in bags has also been studied. The use of IoT in parallel with ICT in communicating the run-time food wastage to the users of a canteen is another idea explored (Koivunen et al., 2020). Current market practices and their impact along with better suggestions is given as well (Spicer, Patel, Bodurtha, & Maslen, 2020). OLIO (<https://olioex.com/>), a food distribution service delivered over handheld device apps, has shared their operating metadata for a study confirming that the relationships developed in the food distribution process contribute in the development of new policies that further promote the use of ICT platforms in this domain (Harvey, Smith, Goulding, & Illodo, 2020). The reverse has also been proven to be true. That is to say, existing policies play a major

role in encouraging or discouraging food distribution as well (Davies, Cretella, & Franck, 2019; Eriksson, Giovannini, & Ghosh, 2020).

Now we discuss some existing work related to Smart Cities, Smart Living, Smart Health, Smart Governance and Sustainability, from various angles including legislative policies, social media reactions on schemes, presence of small businesses, impacts of COVID, relevant app development etc. as our surplus food distribution will circumvent around the possible architecture and implementation of these environments. The international standards that describe a smart city are developed by the International Organization for Standardization (ISO). This is utmost essential in recognizing and promoting smart cities (Lai et al., 2020). These standards prove useful to normalize and benchmark goods and services across smart cities. Study has observed that IoT, Bluetooth and cloud computing are the technologies most discussed in the context of smart cities (Ismagilova, Hughes, Dwivedi, & Raman, 2019). They conclude that several of the United Nations' (UN) sustainable development goals were executable in smart cities. Planning with the people has been considered important and an indicator of good governance. The gainers of smart cities have been arguably the corporations, technology providers and the lobbyists who participate in the planning of the smart cities (Kummitha & Crutzen, 2017). Good governance is the first priority of the residents of these smart cities so as to include them as well in the planning process and listen to their say. What comprises a smart city, its entities, are described in detail as well (Ahad, Paiva, Tripathi, & Feroz, 2020). Smart energy, security, governance, policies, healthcare, living, industry, sustainability are a few of the ones pointed out. The efficacy of smart cities have been narrowed down to the prowess of sensor selection, algorithm selection, and privacy solutions (Leelaarporn et al., 2021). Together, these three can lead the sensor-driven achievement of smart living. The utilization of technology by the citizens play an important role in the smart urbanization (Han & Kim, 2021). The government engaging the citizens in various decision making is considered to be critical in the development of smart cities. The use of technology available in smart cities can create a fabric that supports smart health (Chen et al., 2020). It consists of a network of products embedded with diverse interactive services to form a complete health monitoring and guidance system. Smart living can be achieved as an ecosystem consisting of IoT devices delivering a seamless experience to the citizens (Kor, Yanovsky, Pattinson, & Kharchenko, 2016). However, there are challenges like the integration of different solutions by different providers, and the interoperability between these solutions. The involvement of Artificial Intelligence (AI) in the smart healthcare domain is of particular interest (Kamruzzaman, 2020). This approach provides cost reduction and efficient management of the involved stakeholders, broadening the capability of the healthcare system by miles. Developing countries often miss to address the healthcare concerns of its citizens leading to higher overall mortality rates in such countries (Gahlot, Reddy, & Kumar, 2018). With the proper utilization of the Internet and the IoT platform, smart healthcare will change this statistics soon. Medium to large businesses are moving towards the blockchain based healthcare system from the traditional digital healthcare system (Kumar et al., 2020). The application of blockchain in the healthcare domain has resulted in better security, transparency, immutability and decentralization amongst others. The prerequisites of developing smart and sustainable cities has been declared as advanced technologies and smart governance (Lopes, 2017). It has been noted that the development of smart cities follows the same growth curve as does the development of smart governance. The quality of life of the citizens of a smart city is influenced by the elements of smart governance (De Guimarães, Severo, Júnior, Da Costa, & Salmoria, 2020). Transparency was found to have the highest influence amongst others as it reinforced citizen empowerment by the reduction of corruption. The reaction of the people towards government policies can be mined by the policy makers to enable feedback based smart governance (Puri, Varde, Du, & De Melo, 2018). The COVID-19 pandemic has encouraged the adoption of IoT in several sectors like smart healthcare, smart

home, smart buildings, smart cities, smart transportation, and the Industrial Internet of Things (IIoT) (Umair, Cheema, Cheema, Li, & Lu, 2021). However, device development cost, research grade wearables, easy to use devices, data and architecture standardization, etc. still pose as challenges to the process. Although, smart cities have proved their readiness towards COVID-19 as a result of novel technologies, AI, IoT, big data, and blockchain (Hassankhani, Alidadi, Sharifi, & Azhdari, 2021; Kuzior et al., 2022), the same cannot be said about their responsiveness. With social media inputs as the feedback, decision- and policy-makers can interactively work towards achieving the goals of a smart city (Dahbi, Saadane, & Mbarki, 2019). ML based social media sentiment and event analysis, prediction and classification offer a bottom-up approach towards incremental satisfaction of the citizens of the smart cities (Alam, Abid, Guangpei, & Yunrong, 2020; Hodorog, Petri, & Rezgui, 2022). Although, COVID-19 produces fluctuations, social media analysis has proven to be a reliable source of the citizens perception of the smart city decision making.

The use of ICT based platforms positively speed-up the movement of surplus food across a broader geographical area (Weymes & Davies, 2019). Local food redistribution has been found to produce opportunities as well as pose challenges (Berns & Rossitto, 2019). The high dependency of food value evaluation on the food practices of the region has also been pointed out. As previously discussed, handheld device applications like *OLIO* (<https://olioex.com/>) are very popular today and are a great option for surplus distribution as well (Harvey et al., 2020). This helps increase the speed of communication between the donors and the receivers. Research (Durr & Lotz-Sistika, 2019) on *Food for Us* (<http://foodforus.co.za/>), a South African app that bridge small-scale producers with markets shows that the involvement of technical platforms help mobilize unsold produce. New apps for surplus distribution have been proposed as well (Prova, Rayhan, Shilon, & Khan, 2021). With dynamic user feedback, the barrier between the users and the designers can be removed and it can become another great option for surplus movements. Similarly, web-based operations management systems are also available that reduce the cost and effort of surplus food distribution by automating the movement, thereby making the surplus distribution process more transparent and accessible (Alblihed, Almutairi, Almahmoud, Aladhadh, & Alabdulatif, 2022). The thin line of difference between food and waste is highlighted and the factors that contribute to the decision making process have been pointed out (Spring, Adams, & Hardman, 2019). The dual challenges of food wastage and food unavailability have been shown to be addressed by surplus distribution (Arcuri, 2019). Pointing out at the food wastage laws prevailing in Italy, it has been argued that such laws encourage the adaptive/discounted pricing of expiring food items. Food sharing among peers has been shown to attract a diverse set of actors (King, 2019). Food distribution also creates networks of sharing individuals where the use of ICT based platforms play a major role (Light & Miskelly, 2019). Food distribution in cities is a major field for research and development (Davies & Evans, 2019).

Adaptive pricing for items having wildly fluctuating demand has been proven as a great plan for mitigating wastage (Duan, Cao, & Huo, 2018). The contribution of ICTs in minimizing costs and inefficiencies, and maximizing productivity and coordination leading to a better food movement experience is recognised (El Bilali & Allahyari, 2018). The most benefit of ICT based coordination is pointed out to be received by those perishable food items that have a very short shelf life (Davies & Weymes, 2018). This study also mentions that along with the capability of ICT platforms to cover a geographically wider area comes the moral and social challenges of percolating to diverse localities. This is exactly why surplus distribution gets the nickname “messy” highlighting the fact that the heterogeneity of participants and practices get involved with surplus distributions using ICT based platforms (Rut & Davies, 2018). Multiple food production stages observe the benefits of using ICT based platforms, although, the retail one tops the charts (Weymes

& Davies, 2018). They have also discussed the urban notions for differentiating edible food from waste food and have suggested a few of their own. The rules of the land are as important as all other factors and play a major role in shaping the behaviour of the participants in the act of surplus distribution (Facchini, Iacovidou, Gronow, & Voulvoulis, 2018). The same has been shown to shift us towards a more sustainable economy and get a diverse demography together for the act boosting the economy further (Frigo & Lucchini, 2018). The diverse motives, interactions and the operational effects of surplus distribution were brought to light along with the different options for the participants (Rombach, 2018). Individual food initiatives across several countries were studied with respect to sustainability and were documented as a database (Davies & Legg, 2018).

There is confidence on the reinforcing effects of ICT on surplus distribution and pattern analysis and prediction along with database creation of such activities are already noted (Davies et al., 2017). Surplus distribution along with infrastructure changes in the food production system is believed to be a practical solution towards food waste reduction (Falcone & Imbert, 2017). This approach is expected to counter the increasing demands of food as the population of the world grows. In addition, there is no doubt that the current practices in handling food are nowhere near sustainable in terms of the society, the economy or the environment. In the United States alone, at least 12% people suffer with food insecurity whereas a third of the produce is dumped under the soil (Mousa & Freeland-Graves, 2017). Surplus distribution can counter this effect, albeit with more profound effect in certain areas than others. Not only that, surplus distribution has been shown to have positive impacts on the environment, economy and society (Vittuari et al., 2017). The irony of food wastage and food insecurity coexisting in close proximity in the US has been highlighted (Berry & Acheson, 2017). The opacity of school policies and strategies in handling food wastage has been observed and several factors contributing to the choice of some policies are noted. Traceability of food is an often ignored domain that aids in guaranteed food safety and quality (Ananprakit & Esbjörnsson, 2017). Nonprofit organizations (NPO) improve the value of the surplus by adding fresh produce to it while acting as agents in the movement of the same (Philip, Hod-Ovadia, & Troen, 2017). However, they require added support in terms of infrastructure and regulations to implement this. Surplus distribution is far more preferable than landfills. Although composting comes real close in terms of benefits, both options used together promises the maximum utility for a sustainable future (Fleming, 2017). Shelf-life of perishable items forcing its price to be dynamic is a great policy for the reduction of food waste and also for generating good revenue (Adenso-Díaz, Lozano, & Palacio, 2017; Demirag, Kumar, & Rao, 2017). This has been observed with items having different expiry dates and prices driving buyer behaviour (Herbon, 2017).

Food Tank (<https://foodtank.com/>) points out to the fact that several food sharing communities operate at a global scale. ICT, in simple forms, is used by most of them so as to connect the donors and the receivers, albeit passively over social media (Furbank, 2016). ICT, used in more complex forms, can permeate faster through national borders redistributing food aggressively, utilizing the perishable food window effectively, making seamless communications across the globe, and providing great value in terms of traceability, scalability and reachability.

3. Algorithms and methodology

The flesh and bones of the proposed Modified Food Donor to Receiver Matching with Chronological Acceptance (mFDRM-CA) mechanism is similar to that of the FDRM-CA mechanism (Sanyal et al., 2021), since, the former inherits all the procedures of the latter. The modifications that differentiate the proposed mechanism from the existing one are two folds. Firstly, volunteer assignment for donations has been modified for a better matching performance by the mechanism. Secondly, an improved incentive module has been augmented to the mFDRM-CA matching module to attract higher volunteer participation. We discuss both the alterations in this section.

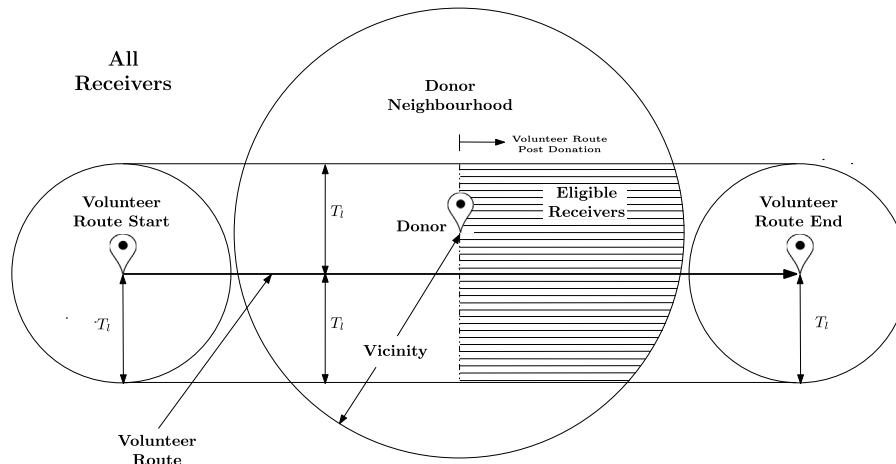


Fig. 1. The mFDRM-CA mechanism considers volunteers having the donor location along their routes before matching optimally.

Table 1

A list of the notations used in the formal representation of the problem.

Notation	Description
$T_l\%$	The original volunteer route distance is the off routing threshold for volunteers.
x	The base amount for incentive calculation for volunteers.
k	The number of activities forming a sliding window for incentive calculation.
c	The maximum joining bonus in dollars (\$) awarded to any volunteer who joins via an invitation link.
n	The number of volunteers in the chain of the delivery person.
L	The activity list of the n volunteers in the aforementioned volunteer chain.
p_i	Denotes participation of volunteer i in the movement of the last k deliveries.
I_i	The incentive that the volunteer in the i th position gets in the chain of n volunteers due to one surplus movement.
I_{Tot}	The total incentive obtained by the n volunteers due to one surplus movement.
I_{Join}	The incentive obtained each by a volunteer and its parent when a newly joined volunteer delivers the first surplus.
u_V	The utility of the volunteer V .
$\hat{u}_V$	The utility of the volunteer V when the volunteer misreports its number of identities.
M	The match produced by the matching mechanism.
$ V $	The total number of volunteers participating in one iteration of the matching process.
$ D $	The total number of donors participating in one iteration of the matching process.
α	The number of duplicate identities of a volunteer appearing in a chain of n volunteers.
β	The multiplier controlling the current availability of volunteers for matching.

3.1. Notations and problem formulation

In Table 1, we present the notations that help our formal representation of the problem. Throughout the paper, two amounts, i.e., x and $2x$, are used extensively and need to be elaborated here. It is assumed that the system has $2x$ fund to distribute for each successful surplus delivery. This is the maximum amount the system distributes for each successful surplus food delivery. For example, suppose a volunteer V has delivered a surplus and $2x = \$4$. In our proposed mechanisms, the volunteer who actually delivers the surplus food will get at most $x = \$2$ and the rest amount out of $2x$ will be distributed in the chain where volunteer V belongs. This amount x is denoted as the base amount for incentive calculations.

We observe that volunteer availability is the limiting factor when it comes to the performance of the matching module. Due to steep fluctuations in the same based on various factors, the matching mechanism

may not be able to produce best fit matches for the available surplus. Our aim is to propose such an incentive mechanism that provides hierarchical incentives so as to encourage volunteer invitation based promotion and participation while discouraging attempts to game the system for personal benefits. We also modify the matching mechanism to optimize volunteer utilization for improved surplus mobility.

3.1.1. The enhanced matching module

In the existing FDRM-CA mechanism, volunteers who have their routes starting close to the donation were considered for assignment to the donor. This was followed by optimizing the choice of volunteers based on several factors such as food type, expiry date, volunteer vehicle details, etc. While the optimization process remains the same with the proposed mechanism, the volunteer pool considered for assignment is enhanced considerably by altering a key condition check. The mFDRM-CA mechanism considers all those volunteers for the donation for whom not only the route start location is close to the donor, but also those volunteers for whom the donor is anywhere along the entire route of the volunteer. This creates a broader pool for donor–volunteer matching as depicted in Fig. 1. Post volunteer assignment, the eligible receivers are selected as in the FDRM-CA mechanism and the volunteer route updated as shown in Fig. 2. This alteration in the volunteer pool consideration incurs constant cost both with respect to time and space but generates far better matches for the agents. This is due to the fact that there are now more volunteers to choose from for each available donation resulting in a higher surplus food mobility.

3.1.2. Incentive calculation

Here, we use the incentive scheme put forward by the MIT team that solved the Defense Advanced Research Project Agency (DARPA) network challenge as discussed in Babaiouff, Dobzinski, Oren, and Zohar (2012) and modify it to fit the recursive nature of our settings where incentives will be provided in multiple rounds. In our scenario, the scheme presented in Babaiouff et al. (2012) cannot be deployed as it is. Therefore, we make some modifications to the incentive scheme to adapt it to our requirements. We first briefly discuss the initial MIT developed scheme followed by the modifications.

The initial MIT developed scheme used incentives to draw in more workforce via crowdsourcing. Fig. 3(a) shows a typical volunteer structure. Volunteers can recruit more volunteers to join the system through them creating a tree like hierarchical structure. For example, in Fig. 3(a), A is the parent of B_1 , B_1 is the parent of B_2 , and B_2 is the parent of C . If C delivers a surplus, C receives $\$x$, B_2 receives $\frac{x}{2}$, B_1 receives $\frac{x}{4}$, and A receives $\frac{x}{8}$. Thus, volunteer payouts, when numbered with 0 for the volunteer who delivers the surplus, to $n - 1$

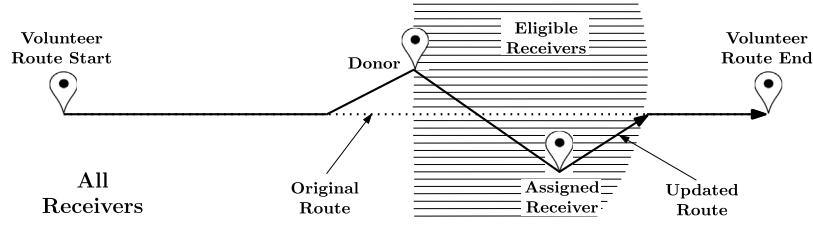
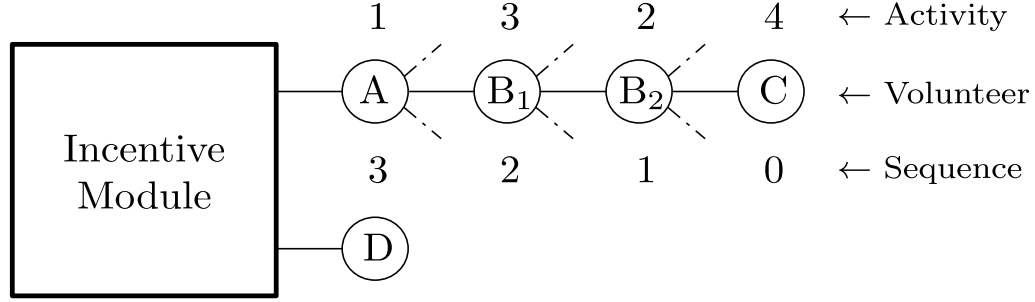


Fig. 2. The updated route of the volunteer post match generation and match acceptance by the involved agents.



(a) Initial structure with multiple volunteer identities.

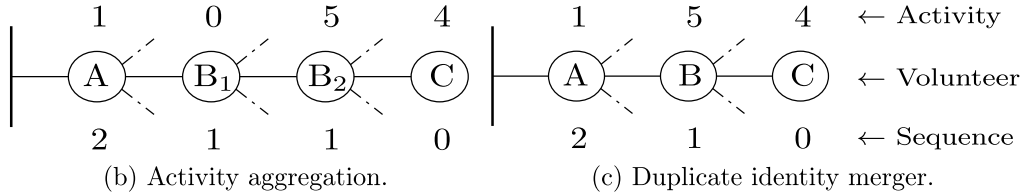


Fig. 3. A typical volunteer arrangement based on the incentive module to demonstrate the module operation.

for the volunteer directly connected to the incentive module, is given by $\frac{x}{2^i}$ for the i th volunteer. We get the total incentive disbursed as:

$$I_{Tot} = \sum_{i=0}^n \frac{x}{2^i} < \sum_{i=0}^{\infty} \frac{x}{2^i} = 2x. \quad (1)$$

However, we have a drawback in this scheme. If a person creates a duplicate identity and invites the second identity from the first one, the person gains. This is the case with A in Fig. 3(a) where B has two identities, B_1 and B_2 , that get $\frac{x}{4}$ and $\frac{x}{2}$, respectively, as incentives. Thus, any person can gain from any number of duplicate identities making the volunteer chains long. There are many modifications to encourage shorter chains. Although effective in reducing the chain length, these modifications effectively negate the most important aim of the crowdsourcing scheme, which is to recruit more volunteers. Herein fits our proposed modifications. These modifications to the initial incentive scheme are two folds. The first one handles duplicate volunteer identities and the second one handles new volunteer joining incentives.

3.1.3. Duplicate identity management

Here we deal with the fact that all volunteers are profit maximizing agents and will create duplicate identities to game the system. To discourage the duplication of identity and also to encourage participation of the volunteers, we introduce a multiplicative factor, *participation*, while calculating volunteer incentives. This *participation* is calculated as follows.

1. Use the last k activities as the sliding activity window for incentive calculation. Here k is a hyperparameter.
2. Compute a_i as the i th volunteer's activity in the last k activities of the invitation chain of the volunteer who performs the surplus delivery.

3. Now, we calculate the participation of each volunteer i towards the last k activities in this invitation chain in either one of the following two ways. We shall discuss the merits of each method later.

- **The Volunteer Oriented Incentive Disbursal (VOID) method:** This method tries to offer maximum benefits to the volunteers. While it penalizes identity duplication, it fails to do so under certain conditions that we will discuss later on. In this method, we calculate the participation of the volunteer as follows:

$$p_i^{VOID} = \frac{a_i}{\sum_{j=0}^i a_j} \quad (2)$$

- **The System Oriented Incentive Disbursal (SOID) method:** This method tries to offer maximum benefits to the system. It always penalizes for identity duplication. In addition, it penalizes increasingly as the number of duplicate identities increase. Participation is calculated for this method as follows:

$$p_i^{SOID} = \frac{a_i}{\sum_{j \in L} a_j} \quad (3)$$

In the given scenario, the incentives that are generated as a result of the VOID scheme are as given in Table 2. Let us take the case of B_2 , who was entitled to an incentive of $\frac{x}{2}$. Here, a sliding window of $k = 15$ activities has been applied. Since, in the invitation chain of C, the total number of activities is less than the sliding window value (i.e., $4 + 2 + 3 + 1 = 10 < 15$), the window value defaults to $k = 10$. Participation of B_2 is calculated as $\frac{2}{4+2} = \frac{1}{3}$. This way, the final incentive of B_2 becomes $\frac{x}{2} \times \frac{1}{3} = \frac{x}{6}$. We continue calculating incentives of other volunteers in a similar way. Note that the participation of both B_2 and B_1 are

Table 2VOID incentives with misreporting volunteers as shown in Fig. 3(a) using $k = 15$.

Seq.	Volunteer	Raw incentive (\$)	Activity	Participation	Final incentive (\$)
0	C	$x/2^0$	4	$\frac{4}{4} = 1$	x
1	B_2	$x/2^1$	2	$\frac{2}{4+2} = 1/3$	$x/6$
2	B_1	$x/2^2$	3	$\frac{3}{4+2+3} = 1/3$	$x/12$
3	A	$x/2^3$	1	$\frac{1}{4+2+3+1} = 1/10$	$x/80$

Table 3VOID incentives after volunteer activity aggregation as shown in Fig. 3(b) using $k = 15$.

Seq.	Volunteer	Raw incentive (\$)	Activity	Participation	Final incentive (\$)
0	C	$x/2^0$	4	$\frac{4}{4} = 1$	x
1	B_2	$x/2^1$	5	$\frac{5}{4+5} = 5/9$	$5x/18$
1	B_1	$x/2^1$	0	$\frac{0}{4+5+0} = 0$	0
2	A	$x/2^2$	1	$\frac{1}{4+5+1} = 1/10$	$x/40$

Table 4SOID incentives with misreporting volunteers as shown in Fig. 3(a) using $k = 15$.

Seq.	Volunteer	Raw incentive (\$)	Activity	Participation	Final incentive (\$)
0	C	$x/2^0$	4	$\frac{4}{4+2+3+1} = 2/5$	$2x/5$
1	B_2	$x/2^1$	2	$\frac{2}{4+2+3+1} = 1/5$	$x/10$
2	B_1	$x/2^2$	3	$\frac{3}{4+2+3+1} = 3/10$	$3x/40$
3	A	$x/2^3$	1	$\frac{1}{4+2+3+1} = 1/10$	$x/80$

the same at $\frac{1}{3}$. Even then their final incentives are different with the volunteer identity having the smaller sequence number getting a higher incentive. This shows that agents with multiple identities will always want to have all their activities in the identity with the lowest sequence number as in 3(b). This is evident from the incentives generated for B with misreporting ($\frac{x}{6} + \frac{x}{12} = \frac{x}{4}$) is less than that after activity aggregation ($\frac{5x}{18}$) as shown in Table 3. That will automatically result in no incentives for the identities having higher sequence numbers and no participation. Furthermore, we merge the identities with zero activities with the immediate neighbour with a lower sequence till an identity with a non-zero activity is reached as in 3(c). This results in the removal of distinct sequence numbers for identities having no participation to ensure that their predecessors are not pushed down the sequence number and deprived of their rightful incentive amount. Therefore, the incentive distribution system is resistant to duplicate identities and encourages greater volunteer recruitment. Later on, we discuss certain special conditions under which this resistance may fail.

A similar situation is observable for the SOID scheme and the incentives resulting from the given scenario is shown in Table 4. The same sliding window of $k = 15$ activities has been used here. Let us check for B_2 who now has a participation of $\frac{2}{4+2+3+1} = \frac{1}{5}$, and gets an incentive of $\frac{x}{10}$. Similarly, B_1 has a participation of $\frac{3}{4+2+3+1} = \frac{3}{10}$, and gets an incentive of $\frac{3x}{40}$. If now, B does all the activities in B_2 , then B_2 has a new participation of $\frac{5}{4+2+3+1} = \frac{1}{2}$, and gets an incentive of $\frac{x}{4}$ as in Table 5. We see that this incentive for honesty ($\frac{x}{4}$) is more than that of identity duplication ($\frac{x}{10} + \frac{3x}{40} = \frac{7x}{40}$). Thus, volunteer B gets motivated to be honest. Also, since in both the schemes the sequence number of A changes to 2, A gets double the incentive amount in this setup when B is honest as that when B is not. In general, when all volunteers are honest, the incentive that any predecessor of dishonest volunteers get is $I_{i+1} = 2^d \times \frac{x}{2^{i+1}}$ where d is the total number of duplicate identities of the dishonest volunteers.

Our proposed SOID mechanism is to make sure that the volunteers cannot manipulate the chain. This will ensure three facts: 1. The system is already in equilibrium, 2. It is easy for the volunteer to participate in the system by recruiting truthfully, 3. If everybody manipulates,

the system will be in a chaos. However, as the volunteers still can recruit others (truthfully), they can have the benefits by recruiting others, and thereby can influence the chain. In this paper, another mechanism is put forward which is VOID, and in this mechanism, though the volunteers can gain by self duplication, but finding that beneficial arrangements they need to do much more mental calculations. In the VOID mechanism, the volunteers can influence the chain by recruiting others truthfully as well as by self duplication. Here the SOID mechanism is platform centric (saving cost for the platform), and VOID mechanism is little bit oriented towards volunteers (as there is a possibility of self duplication). We present both the mechanisms and leave the choice of selecting one of them to the implementation environment requirements.

3.1.4. Budget feasibility

From Eq. (1), we know that the incentive scheme used in the DARPA red balloon challenge (Babaioff et al., 2012) disbursed a maximum total incentive of $2x$, where x is the base amount for incentive calculation. We compare the VOID and the SOID mechanisms against this scheme to prove its budget feasibility.

Lemma 1. Total incentive disbursed by the VOID mechanism is always less than $2x$ where x is the base amount for incentive calculation.

Proof. The total incentive disbursed by the VOID mechanism is:

$$\begin{aligned}
 I_{Tot}^{VOID} &= \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^i a_j} = \frac{x}{2^0} \times \frac{a_0}{\sum_{j=0}^0 a_j} + \sum_{i=1}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^i a_j}, \\
 &< x + \sum_{i=1}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^i a_j} = x + x \times \sum_{i=1}^n \frac{1}{2^i} \times \frac{a_i}{\sum_{j=0}^i a_j}, \\
 &< x + x = 2x.
 \end{aligned} \tag{4}$$

Thus, we see that the total incentive that is disbursed using the VOID mechanism, namely, I_{Tot}^{VOID} , is always less than $2x$. $\square$

Lemma 2. Total incentive disbursed by the SOID mechanism is always less than $2x$ where x is the base amount for incentive calculation.

Table 5

SOID incentives after volunteer activity aggregation as shown in Fig. 3(b) using $k = 15$.

Seq.	Volunteer	Raw incentive (\$)	Activity	Participation	Final incentive (\$)
0	C	$x/2^0$	4	$\frac{4}{4+5 \times 0+1} = 2/5$	$2x/5$
1	B_2	$x/2^1$	5	$\frac{4}{4+5 \times 0+1} = 1/2$	$x/4$
1	B_1	$x/2^1$	0	$\frac{0}{4+5 \times 0+1} = 0$	0
2	A	$x/2^2$	1	$\frac{1}{4+5 \times 0+1} = 1/10$	$x/40$

Proof. For this proof we can argue in similar lines as the proof for the VOID mechanism. The total incentive disbursed in the SOID mechanism will be:

$$\begin{aligned}
 I_{Tot}^{SOID} &= \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j} = \frac{x}{2^0} \times \frac{a_0}{\sum_{j=0}^n a_j} + \sum_{i=1}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j}, \\
 &< x + \sum_{i=1}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j} = x + x \times \sum_{i=1}^n \frac{1}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j}, \\
 &< x + x = 2x.
 \end{aligned} \quad (5)$$

Therefore, the total incentive that is disbursed using the SOID mechanism, I_{Tot}^{SOID} , is also always less than $2x$. $\square$

3.1.5. Word of mouth encouragement

Here, we address the issue that new volunteers may wish to bypass the joining link sent to them by other volunteers and decide to join the system directly, robbing their inviting volunteers of their incentives. We here distribute an one time joining incentive when a new volunteer joins through an invitation from another volunteer. This incentive is calculated as follows.

1. The incentive is generated and distributed on the first activity of the new volunteer.
2. The incentive amount per volunteer is capped at $\$c$, another hyperparameter.
3. The total incentive amount is distributed equally between the parent and the child volunteers.
4. The total incentive amount is the difference between the maximum cumulative possible incentive for one activity ($\$2x$) and the actual cumulative incentive distributed. For the two incentive disbursal methods this value will be different.

- **For the VOID mechanism:** Total joining incentive is

$$I_{Join}^{VOID} = \sum_{i=0}^{\infty} \frac{x}{2^i} - \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j} = 2x - \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^n a_j}. \quad (6)$$

- **For the SOID mechanism:** Total joining incentive is

$$I_{Join}^{SOID} = \sum_{i=0}^{\infty} \frac{x}{2^i} - \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j \in L} a_j} = 2x - \sum_{i=0}^n \frac{x}{2^i} \times \frac{a_i}{\sum_{j \in L} a_j}. \quad (7)$$

Thus, both the parent and the child receive half of the total joining incentive amount. This ensures that both the volunteers are interested in volunteer invitations instead of going around and joining the system directly. In addition, since the incentive is generated and distributed on the first activity of the child volunteer, it ensures that the new volunteer is active after joining. The capping amount ensures that there is still scope for charity, as in the initial scheme by the MIT students, or savings, as the requirement may be.

Lemma 3. Working in a chain is always more profitable than working individually.

Proof. A volunteer V can join the incentive scheme in any of the positions directly attached to the system out of A , C or D as shown

in Fig. 3. To prove this lemma for V , we define the utility function of the volunteer as follows:

$$\begin{aligned}
 u_V &= u_{individual} + u_{chain} = p_0 \times x + \sum_{i>0} p_i \times \frac{x}{2^i}, \\
 &= (p_0 + d) \times x, \text{ where } d = \sum_{i>0} p_i \times \frac{1}{2^i}.
 \end{aligned} \quad (8)$$

Here, p_0 is the participation of volunteer V when working individually, and p_i ($i > 0$) is the participation for V when the volunteers in the invited chain of V work individually. If V does not invite any volunteer and decides to work individually, then V has utility defined by $u_V = p_0 \times x$ for each activity. Whereas, if V decides to invite more volunteers and work in a chain, the new utility is defined by $u_V = (p_0 + d) \times x$ which is more than $p_0 \times x$ for $d > 0$ accounting for the activities of the invited volunteers as well. Thus, it is evident that a volunteers makes more profit working in a chain than individually. $\square$

Observation 1. For the volunteers, an app will be calculating the incentives for them. The volunteers just need to deliver the food items and, according to their wish, they can recruit other volunteers. Other things will be taken care of by the system. It is to be noted that, explanations of the two mechanisms, the VOID and the SOID, are given in the examples in Tables 2, 3, 4, and 5 of Section 3.1.3. From the explanations the volunteers will understand that when they are in a chain and when they are actually supplying food, both are beneficial to them. With this knowledge, they will participate in the system and the system will calculate the incentives and pay them back.

3.2. Proposed mechanism

The Modified Food Donor to Receiver Matching with Chronological Acceptance (mFDRM-CA) mechanism, has two distinct modifications to enhance its efficiency. One is while the algorithm checks for volunteers for a donation. This enables volunteers to be assigned donation pickups from any location along their route instead of only from near their start location as previously. This affects only a condition check in the algorithm and is therefore omitted here. Although a minor change, it greatly affects the efficiency of the matching process by availing the donors a larger pool of volunteers from which to choose. The other modification is the addition of an improved incentive module so as to attract a higher volunteer participation. As previously, this module works with the matching module to stabilize the availability of volunteers. The algorithms for these methods are provided in this section.

The matching algorithm (Algorithm 1) does three things in parallel. Firstly, it calls the incentive module. Secondly, it starts collecting agent requests in parallel. Thirdly, it starts matching the available requests in parallel. In this third parallel work, it retrieves available donors and receivers and updates their preferences according to the availability of the agents. Thereafter, each donor is selected for assigning volunteers. All volunteers who have this donor close to their route up to a maximum off routing of $T_1\%$ of their total travel distance are pooled together. This volunteer pool selection is the pivotal change that makes this algorithm perform far better than its predecessor. This performance upgrade is glaringly visible under restricted volunteer availability. The best fit volunteer is then selected for this donor from that pool as per the

donation shelf life and the available receiver locations. Now, for each receiver meal request, a donor meal is assigned as per the availability.¹

Algorithm 2 (the VOID mechanism) and Algorithm 3 (the SOID mechanism) are incentive modules. Both work by calculating the activities of volunteers in the volunteer chain with a successful delivery. This activity is calculated in a moving window of k activities. The positions of the volunteers start with 0 for the one who delivers the surplus, and ends with $n - 1$ for the one directly attached to the incentive module as in Fig. 3(a). Each volunteer is assigned a proportionate incentive as per the mechanism calling it (VOID or SOID) before calling Algorithm 4 for calculating the joining bonus.

Algorithm 4 checks whether this delivery is the first surplus delivery for the 0th volunteer. If so, it provides the 0th volunteer, and its parent, the 1th volunteer with the joining bonus.

Algorithm 1: mFDRM-CA()

```

1 do in parallel // Call incentive module.
2   call VOID( $x, c, n, k, L$ ) / SOID( $x, c, n, k, L$ ).
3 do in parallel
4   Collect fresh agent requests.
5 do in parallel // Match requests.
6   foreach request do
7     Sort requests into perishable and non-perishable.
8   Get available donors.
9   Get available receivers.
10  Update donor preferences.
11  Update receiver preferences.
12  foreach donor do
13    Get available volunteers having this donor around the
    route up to a maximum tolerance of  $T_l\%$  of the total
    travel distance of the volunteer.
14    Assign volunteer by optimizing transportation
    requirements.
15  foreach receiver do
16    foreach requested meal do
17      Match meal request with available meal from
      donor.
18  return Matches

```

Algorithm 2: VOID(x, c, n, k, L)

```

1  $I \leftarrow \Phi$  // Initialize incentive array for  $n$  volunteers.
2  $totAct \leftarrow 0$  // Initialize activity counter.
3  $totIncentive \leftarrow 0$  // Initialize total incentive.
4  $A \leftarrow \text{sort}(L)$  // Sort activities reverse chronologically.
5  $cutoffTime \leftarrow A[k].Timestamp$  // Get timestamp of  $k^{th}$  activity.
6 for  $i \leftarrow 0$  to  $n - 1$  do // Each volunteer from delivery agent to root.
7    $actWin \leftarrow 0$  // Initialize sliding window volunteer activity.
8   foreach  $activity \in A[i]$  do // Activity is in last  $k$  activities.
9     if  $activity.Timestamp \geq cutoffTime$  then
10       $actWin = actWin + 1$  // Volunteer activity count.
11    $totAct = totAct + actWin$  // Total activity in  $k$  activity window.
12    $I[i] = I[i] + \frac{x}{2^i} \times \frac{actWin}{totAct}$  // Incentive generated for  $i^{th}$  agent.
13    $totIncentive = totIncentive + I[i]$  // Total incentive disbursed.
14  $J \leftarrow 2x - totIncentive$  // Amount for joining bonus.
15 call JOIN( $c, A, I, J$ ) // Calculate volunteer joining bonus.

```

¹ For finer grained details on the matching module, please refer to our previous work (Sanyal et al., 2021).

Algorithm 3: SOID(x, c, n, k, L)

```

1  $I \leftarrow \Phi$  // Initialize incentive array for  $n$  volunteers.
2  $totIncentive \leftarrow 0$  // Initialize total incentive.
3  $A \leftarrow \text{sort}(L)$  // Sort activities reverse chronologically.
4  $cutoffTime \leftarrow A[k].Timestamp$  // Get timestamp of  $k^{th}$  activity.
5  $totAct \leftarrow \sum_{cutoffTime} A$  // Initialize activity counter.
6 for  $i \leftarrow 0$  to  $n - 1$  do // Each volunteer from delivery agent to root.
7    $actWin \leftarrow 0$  // Initialize sliding window volunteer activity.
8   foreach  $activity \in A[i]$  do // Activity is in last  $k$  activities.
9     if  $activity.Timestamp \geq cutoffTime$  then
10       $actWin = actWin + 1$  // Volunteer activity count.
11    $I[i] = I[i] + \frac{x}{2^i} \times \frac{actWin}{totAct}$  // Incentive generated for  $i^{th}$  agent.
12    $totIncentive = totIncentive + I[i]$  // Total incentive disbursed.
13  $J \leftarrow 2x - totIncentive$  // Amount for joining bonus.
14 call JOIN( $c, A, I, J$ ) // Calculate volunteer joining bonus.

```

Algorithm 4: JOIN(c, A, I, J)

```

1 if  $A[0] == 1$  then // New user incentive.
2    $joinBonus = \min(J/2, c)$  // Joining bonus.
3    $I[0] = I[0] + joinBonus$  // Child volunteer.
4    $I[1] = I[1] + joinBonus$  // Parent volunteer.
5 return  $I$  // Volunteer incentive amount.

```

4. Algorithm analysis

We present the analysis and the various desirable properties of the mFDRM-CA matching and incentive modules in this section.

4.1. Correctness

Lemma 4. *mFDRM-CA works correctly.*

Proof. The proposed mFDRM-CA mechanism inherits all the algorithms of the FDRM-CA algorithm (Sanyal et al., 2021) and due to this similarity, the correctness proofs remain the same for the mFDRM-CA mechanism as well. Thus, to prove the correctness of the proposed mechanism, we need to only prove the correctness of the incentive module using both the VOID and the SOID mechanisms, we use the loop invariant technique (Cormen, Leiserson, Rivest, & Stein, 2009). The following loop invariant is used:

At the beginning of the i th iteration, the $(i-1)$ th volunteer's incentive is already calculated.

Initialization: Before the first iteration of the loop, there are no volunteers and the invariant is trivially satisfied.

Maintenance: At any iteration i , the general incentive and the joining bonus is calculated for the i th volunteer. Thus, at the beginning of the next iteration the invariant is maintained.

Termination: At termination, when $i = 0$, each of the n volunteers have their incentives calculated upholding the invariant. This proves that mFDRM-CA works correctly. $\square$

4.2. Strategyproofness

Definition 1. *Strategyproofness* under these settings is defined as the fact that the volunteers can never gain by misreporting. That is, even if a volunteer creates multiple identities to gain incentives on them, the total incentive on all the identities taken together will always be less than the incentive that the volunteer would get on a single identity doing the same amount of activity without identity duplication. For a volunteer B with a unique identity, the utility function is given

by $u_B = u_{\text{individual}} + u_{\text{chain}}$. If the volunteer B misreports and creates multiple identities, the utility of each of these identities is given by the generalized equation for the l th identity as $\hat{u}_B^l = u_{\text{individual}}^l + u_{\text{chain}}^l$. Thus, the combined utility of the volunteer B using multiple identities is given by $\hat{u}_B = \sum_l \hat{u}_B^l$. For strategyproofness condition, we need to have $\hat{u}_B \leq u_B$.

Lemma 5. *mFDRM-CA is strategyproof for all agents using the SOID mechanism.*

Proof. While the matching module is already proved to be strategyproof for donors and receivers (Sanyal et al., 2021) and the incentive module has nothing to do with it, all we are left to prove is that the incentive module is strategyproof for volunteers while the SOID mechanism is used. To game the system, volunteers can resort to using multiple identities (IDs) inviting one from the other before recruiting new volunteers. This only results in the distribution of activities of the defaulting volunteers over their different IDs. We already have the utility of the volunteer B in Fig. 3 as $u_B = u_{\text{individual}} + u_{\text{chain}}$, where $u_{\text{individual}} = x \times p_0$ is the amount for the volunteer's individual activity. Since, our incentive scheme is dependent on volunteer activity at each ID, it multiplies the raw incentive with a factor derived from the activity count of the ID. We here focus on the volunteer's variable incentive that arises from the activity of the chain given by $u_{\text{chain}} = \sum_{i>0} p_i \times \frac{x}{2^i}$. Here, i (> 0) is the sequence number taking different values for different activities by the volunteers invited by B . Here, we have B making multiple identities B_1, B_2 and has invited B_2 from B_1 . Now B distributes $t > 0$ ($t = t_1 + t_2$) tasks among these two identities with t_1 tasks done in B_1 and t_2 tasks done in B_2 . Assuming that C did a successful delivery, the rewards of B_1 and B_2 will be given by:

$$u_{B_2} = \frac{t_2}{a_c + t} \times \frac{x}{2^1}, \text{ and, } u_{B_1} = \frac{t_1}{a_c + t} \times \frac{x}{2^2}. \quad (9)$$

where a_c is the activity of volunteer C in the k activity sliding window. The truthful reward for B would be:

$$u_B = \frac{t}{a_c + t} \times \frac{x}{2^1}. \quad (10)$$

Adding the corresponding components of Equation set (9), we get the incentive that B gets by being dishonest as:

$$\begin{aligned} \hat{u}_B &= \frac{t_2}{a_c + t} \times \frac{x}{2^1} + \frac{t_1}{a_c + t} \times \frac{x}{2^2} = \frac{1}{2} \times \frac{t + t_2}{(a_c + t)} \times \frac{x}{2^1}, \\ &= \frac{1}{2} \times \frac{t}{a_c + t} \times \frac{x}{2^1} + \frac{1}{2} \times \frac{t_2}{a_c + t} \times \frac{x}{2^1} = \frac{1}{2} \times u_B + \frac{1}{2} \times \frac{t_2}{a_c + t} \times \frac{x}{2^1}, \\ &< \frac{1}{2} \times u_B + \frac{1}{2} \times u_B = u_B. \end{aligned}$$

This means that while the volunteer would have gotten a greater incentive (u_B) with all activities on the identity that joined latest, now the volunteer receives a smaller incentive ($\hat{u}_B$) due to distributing the activities over multiple IDs. Thus, a misreport can only result in a loss, making mFDRM-CA used with the SOID mechanism strategyproof for volunteers, and thereby all agents. $\square$

4.3. Pareto-optimality

Lemma 6. *mFDRM-CA is Pareto-optimal for donors.*

Proof. We use mathematical induction for this proof. Let us assume that mFDRM-CA is not Pareto-optimal and that there exists another algorithm OPT that matches agents Pareto-optimally. The base condition is trivially satisfied for zero iterations. Now in the i th induction step, say both the algorithms have performed equally until now and the matching produced by them till this step is the same and is given by M_i . In the $(i+1)$ th step, mFDRM-CA matches the available donors with the most preferable receivers amongst those still available for matching and produces M_{i+1} . If the OPT mechanism produces any matching in this

step that is different from M_{i+1} , it is not Pareto-optimal. Thus, by the Principle of Mathematical Induction, the final match M^{OPT} will not be Pareto-optimal. This is a contradiction to our assumption. Therefore, our initial assumption that mFDRM-CA is not Pareto-optimal is not true. In other words, mFDRM-CA produces Pareto-optimal matching $M^{mFDRM-CA}$. $\square$

4.4. Real-time operation

Lemma 7. *mFDRM-CA produces real time results.*

Proof. It is intuitive that the proposed mFDRM-CA matching module has the same time complexity as that of the unmodified one. This happens due to the fact that we replaced the volunteer pool selection process with a better one having the same constant time complexity. However, the proposed mechanism has two modules working simultaneously. To calculate the time complexity of the whole mechanism we have the following to consider:

- The matching module takes $O(n^3)$ as previously discussed (Sanyal et al., 2021).
- The incentive module takes $O(kn) = O(n)$ as k is a small constant ($k \ll n$).

where n is the number of volunteer agents and k is the number of activities for the sliding window calculation of incentives. Thus, the time complexity is capped at $O(n^3)$ making it work in real time. $\square$

5. Results

Figs. 4, 5 and 6 present the simulation results of the proposed mechanism against the existing mechanism. Fig. 4 contrasts the incentives of a volunteer with multiple identities against a single identity when using the VOID mechanism for incentive disbursal. Similarly, Fig. 5 contrasts the incentives of a volunteer with multiple identities against a single identity when using the SOID mechanism for incentive disbursal. In both Figs. 4 and 5 the X -axis represents the *activity pattern* which is the pattern formed by the numbers of activities performed by the n volunteers of the volunteer invitation chain in the k -activity window. In Fig. 6, volunteer and receiver availability have been expressed in terms of multiples of donor availability ($\times \text{Donors}$). Fig. 6 contrasts the matching modules of the proposed and the existing modules while varying the availability of volunteers.

5.1. Setup

For the simulation, we have used real data taken from a real life smart city context, namely, we have used the OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us> and cast it to our simulation settings using Python libraries. The simulations were carried out for 200 donations and proportionate receivers and volunteers as relevant for each graph. Operation territory was set to 500×500 km. An 18 h operation window was taken from 06:00 h to 23:59 h. Both the incentive mechanisms, namely, the VOID and the SOID, were simulated with different levels of identity duplication. Other simulation parameters were kept similar to that of our previous work (Sanyal et al., 2021). For the VOID mechanism simulation as in Fig. 4, and the SOID mechanism simulation shown in Fig. 5, we use the following equation set for calculating the total incentive of the volunteer having α identities in the system chained to each other.

$$I_{Tot}^{VOID} = \sum_{i=1}^{\alpha} \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^i a_j}, \text{ and, } I_{Tot}^{SOID} = \sum_{i=1}^{\alpha} \frac{x}{2^i} \times \frac{a_i}{\sum_{j=0}^{n-1} a_j}. \quad (11)$$

In the matching module, the availability of volunteers that rule the matching process is given by $|V| = \beta \times |D|$ where β is the multiplier controlling the current availability of volunteers for matching. Here

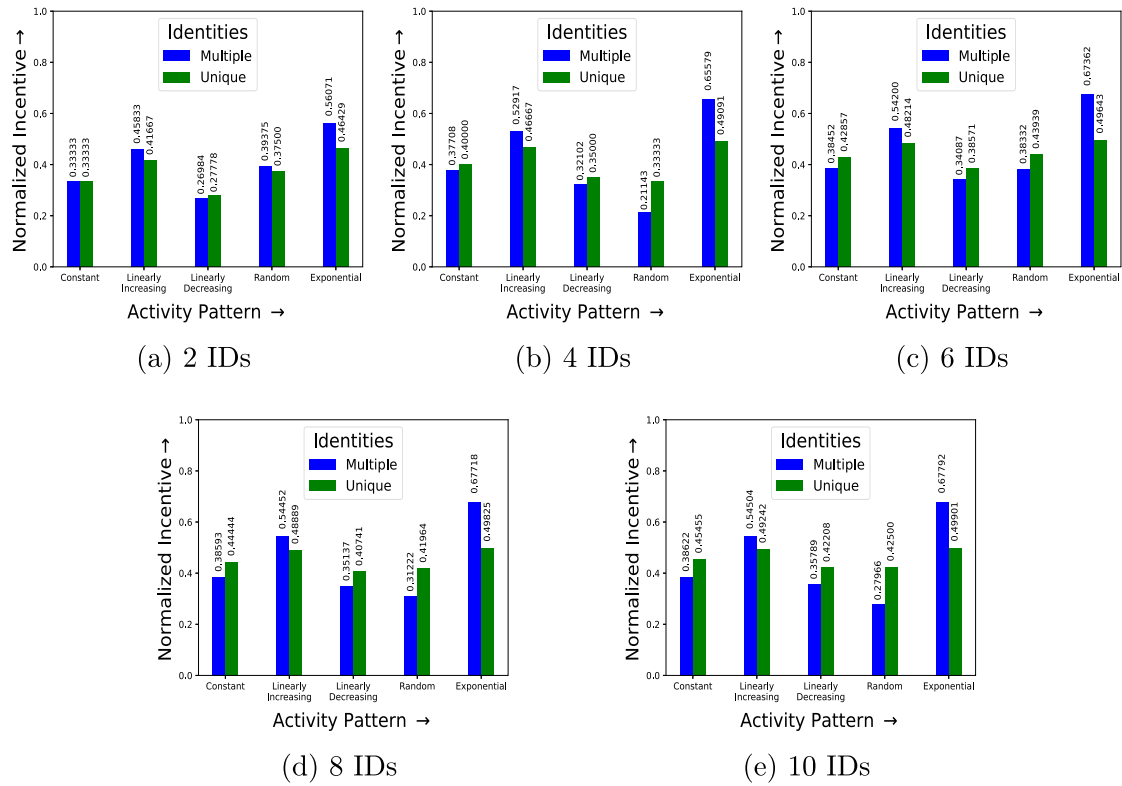


Fig. 4. Contrasting the incentives of a volunteer with multiple identities for the VOID mechanism.

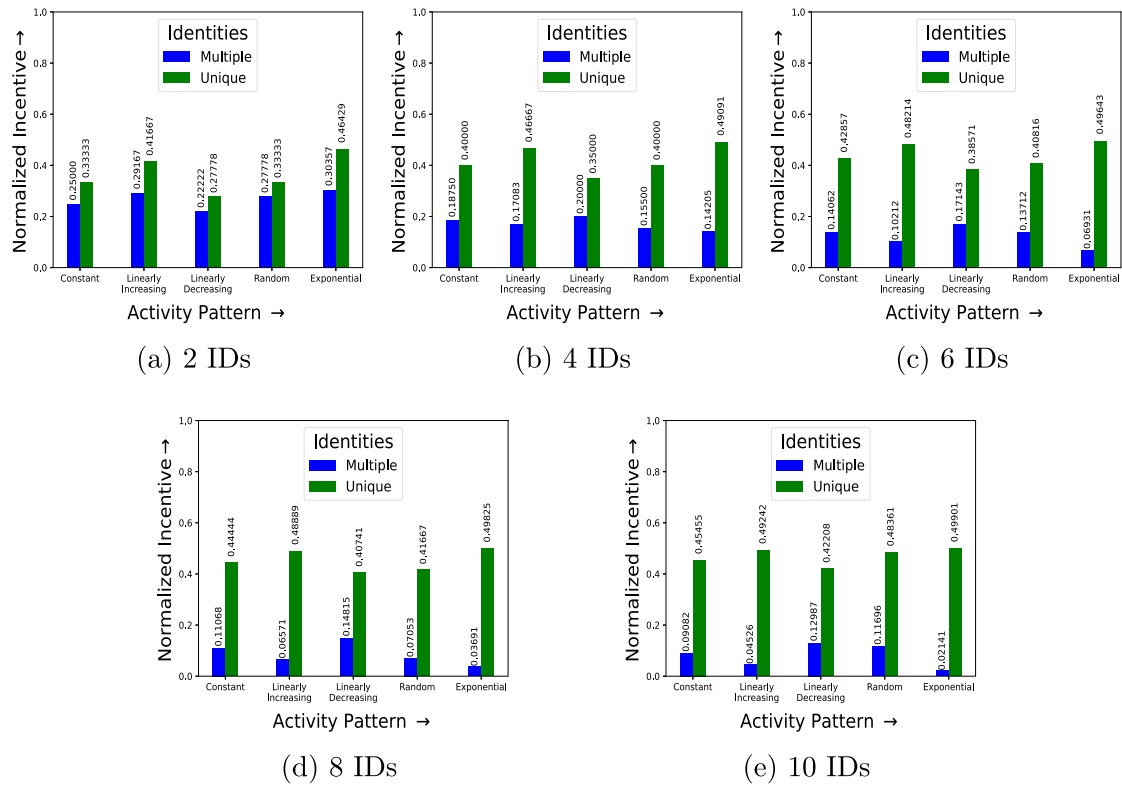


Fig. 5. Contrasting the incentives of a volunteer with multiple identities for the SOID mechanism.

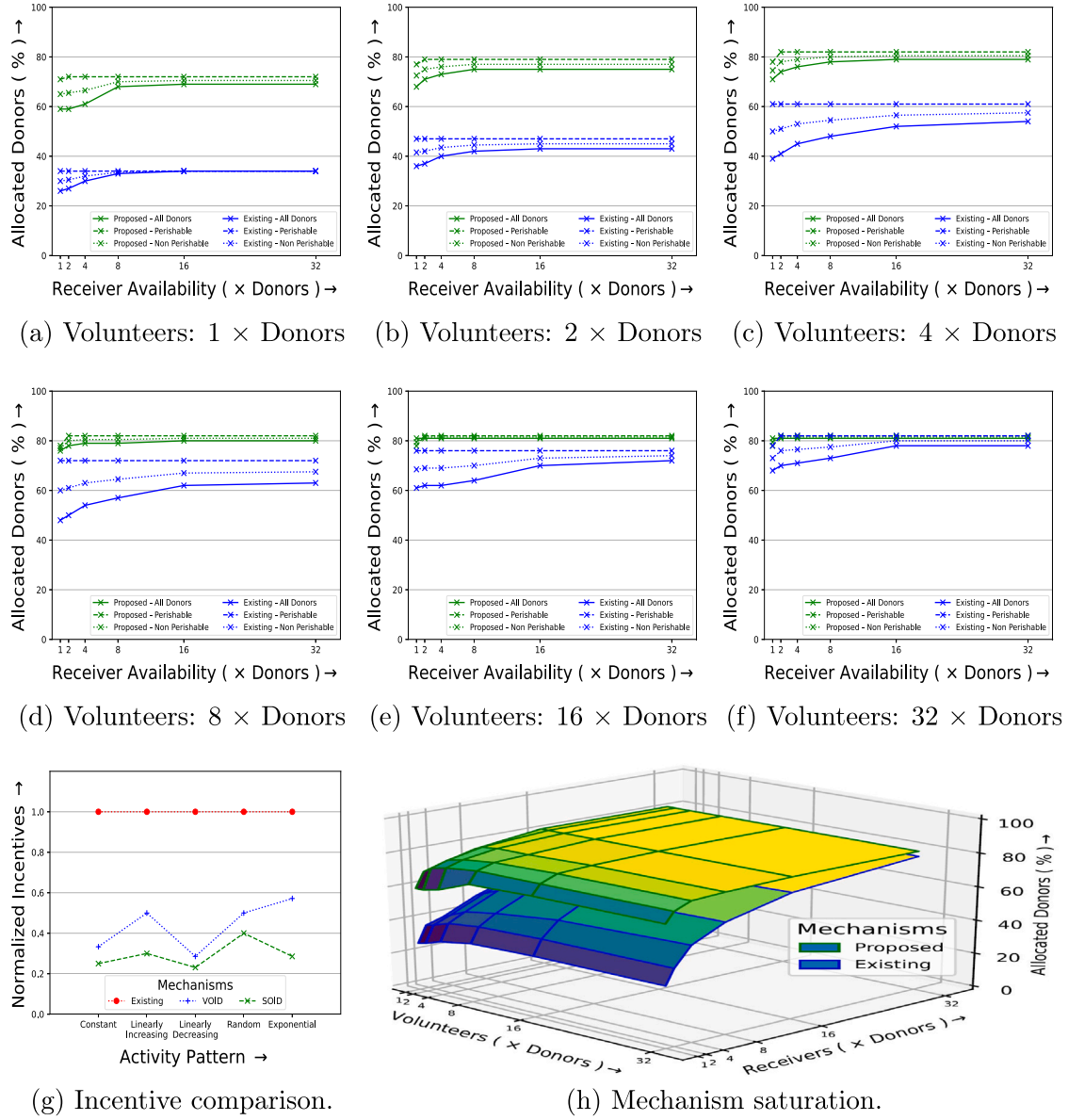


Fig. 6. Simulation results comparing the performances of the existing and the proposed matching mechanisms.

$|V|$ and $|D|$ are the numbers of available volunteers and donors, respectively. Under these settings we evaluate our proposed mechanism against the existing one by varying the values of $\alpha = \{2, 4, 6, 8, 10\}$ and $\beta = \{1, 2, 4, 8, 16, 32\}$.

5.2. Volunteer altruism

In this paper, we assumed that when the volunteers are to be motivated by the incentives, then how the mechanisms will behave. In Geraskin (2021), a propensity measure of altruism is considered that may include the volunteers who are initially selfless and provide voluntary service. Though our *recruitment* based mechanisms are slightly different than Geraskin (2021), we can include that philosophy of propensity measure which will help us incorporate both altruistic and non-altruistic behaviours of the volunteers at one place. By following the idea of the propensity measure to altruism proposed in Geraskin (2021), we have introduced this portion where we have put forward how both altruistic and non-altruistic behaviours of the volunteers can be incorporated in our proposed mechanisms (VOID and SOID). In Fig. 7(a) it is shown that, in this modified setup, both altruistic and

non-altruistic volunteers are present in our chain. In Fig. 7(b), it is shown that if we increase the incentives then the altruistic chain is reduced and the non-altruistic chain is increased. This is because, upto a certain level of incentive threshold the altruistic volunteers will be still inclined to provide free services and beyond that some of the altruistic volunteers will be shifting themselves to non-altruistic chains. The number of such shifts (from altruistic to non-altruistic) is captured by the following equation:

$$n^* = m^* + (1 - \delta)\bar{m}^*.$$

where m^* is the number of volunteers in the non-altruistic chains, $\bar{m}^*$ is the number of volunteers present in the altruistic chains, n^* is the total number of volunteers present in the system, δ is the probability that the volunteers from an altruistic chain will not shift to a non-altruistic chain based on the increase in incentives. If we have $\delta = 1$, then the implication is that the incentives are low and the volunteers are not shifting. So, in this case, the incentives will be disbursed to $n^* = m^* + (1 - 1)\bar{m}^* = m^*$ number of volunteers. Otherwise, if $0 < \delta < 1$, then it will exhibit the case shown in Fig. 7(b) where some altruistic volunteers will be shifted to the non-altruistic chains.

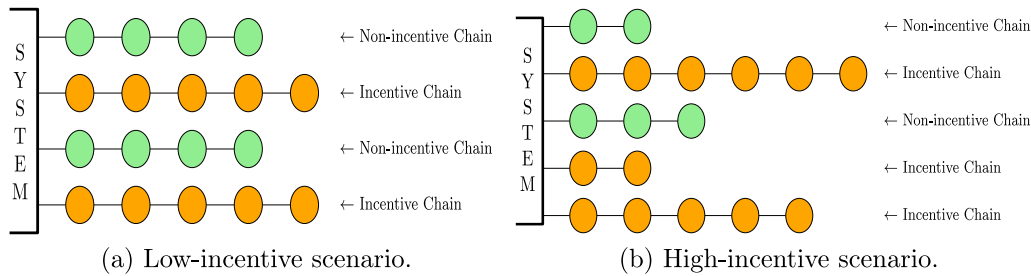


Fig. 7. Shift in altruism of the volunteers as the incentive amount increases. The volunteers (represented by the circles) are assumed to be attached to the incentive system in a hierarchical structure in either purely altruistic or purely non-altruistic chains. Incentive chain is for non-altruistic volunteers, and non-incentive chain is for altruistic volunteers.

5.3. Analysis

Fig. 4 presents the incentives that the VOID mechanism offers to a volunteer through multiple identities against that obtained with a single identity. Fig. 5 presents the incentives under the same settings using the SOID mechanism. It is to be observed that while a volunteer can never gain using multiple identities in the SOID mechanism, it is not the same for the VOID mechanism. In fact, it can be observed that for *linearly increasing* and *exponentially increasing* as well as occasionally for the *random* activity patterns, multiple identities provide more incentive to the volunteer. However, the SOID mechanism never lets the volunteer gain by using multiple identities. As a matter of fact, it penalizes the volunteer increasingly with increasing number of duplicate identities.

Fig. 6(a) through Fig. 6(f) contrast the effects of the availability of volunteers and receivers on the performance of the proposed and the existing mechanisms. Although the existing mechanism performs nearly as well when volunteer availability is very high, it is evident from these results that the proposed mechanism performs far better at lower availability of volunteers. The proposed mechanism also maintains similar allocations of agents in the perishable and non perishable food categories. Perishable food items still get a higher precedence over the non perishable ones when matching.

Fig. 6(g) presents a comparison between the incentives distributed to a random volunteer by the existing and the two proposed incentive mechanisms. It can be observed that both the proposed mechanisms provide cost savings over the existing mechanism, with the SOID mechanism providing the maximum savings to the system. This is because they only distribute a fraction of the incentive that the existing mechanism distributes. The sustainability of the platform will be better maintained if we have these cost saving mechanisms as the system can have more rounds performed.

Fig. 6(h) presents the speed with which the two mechanisms reach saturation in terms of matching the agents for the generated agent request data. We observe that the proposed mechanism reaches its maximum matching potential earlier than the existing mechanism. This clearly represents the potential of the proposed mechanism to perform better under lower volunteer and receiver concentrations. In addition, the inclusion of the incentive module results in an increase in the availability of volunteers. This in turn shifts the current availability of volunteers to the higher end of the scale further increasing the matching performance of the mechanism.

6. Discussion

The above results confirm the following claims presented in this paper:

- The mFDRM-CA matching module shows better performance due to the fact that the matching algorithm assigns volunteers to donors along the route in addition to donors lose to the start location of the route.

- The mFDRM-CA incentive module attracts more volunteer participation tipping low volunteer availability scenarios to higher ones resulting in better match generation performance.
- The mFDRM-CA, using the SOID mechanism, is strategyproof for the volunteers and encourages higher volunteer activity as well by rewarding volunteers having higher activity with higher incentives.
- The mFDRM-CA stabilizes the availability of volunteers over time through the disbursal of incentives. As this is the greatest factor in improving surplus mobility, it contributes to the better performance of the mechanism.

As our incentive scheme for surplus food distribution is motivated by a real life problem (red balloon contest) posed by DARPA, it may be worthy to show that our incentive schemes also work for that real life problem. This will fulfil our desired objective that in a budget constrained environment more rounds can be performed as the system can save some money in each round exploiting the case that the agents may not manipulate the system. This is explored through the scenario presented in Fig. 3(c). The incentives generated by the existing and the two proposed (VOID and SOID) mechanisms for this case is given in Table 6. We can see that the incentive which is being paid to the volunteers by the existing mechanism is the highest amongst all three mechanisms, which does not meet our desired objectives (needing more rounds and more surplus deliveries). The proposed VOID mechanism reduces the incentives of all the volunteers other than the 0th volunteer. The proposed SOID mechanism reduces the incentives of all the volunteers further. Thus, the proposed two mechanisms provide cost savings to the system over the existing mechanism, proving as attractive options for the system administrator.

7. Conclusion

Surplus food distribution is one of the best available solutions to the global menace that is wastage of edible food. The throwing away of the portion of food that is perfectly consumable is countered by surplus distribution either through individuals or through communities. ICT and IoT boost the performance of such food movement activities by making high speed coordination possible among the involved agents. The existing mechanism provides a base for individuals and startups in this area to begin with their operations. However, it suffers with performance issues under lower volunteer availability. To complicate the situation further, the availability of volunteer is shown to be fluctuating rapidly.

The proposed mFDRM-CA mechanism offers a solution to these challenges by modifying the matching mechanism and introducing an incentive module to inspire higher volunteer participation. The findings of our paper could be summarized as follows:

- The proposed matching mechanism produces better matches even during lower volunteer availability,
- The proposed incentive mechanisms provide higher cost savings to the system as compared to the existing one,

Table 6

Comparison of incentives generated by the existing and the two proposed mechanisms (VOID and SOID) presented in § for the volunteer hierarchy shown in Fig. 3(c) using $k = 15$.

Seq.	Volunteer	Existing mechanism	Activity	VOID mechanism	SOID mechanism
0	C	$x/2^0 = x$	4	x	$4x/10$
1	B_2	$x/2^1 = x/2$	5	$5x/18$	$5x/20$
2	A	$x/2^2 = x/4$	1	$x/40$	$x/40$

- The proposed incentive mechanisms encourage active volunteers and penalize inactive ones,
- The proposed mechanisms contribute towards making the system more sustainable in the long run,
- The proposed mechanisms are validated by simulation using real life data.

The modification in the matching process allows for a greater volunteer pool per donation resulting in a higher overall surplus mobility. The incentive module generates incentive for each successful volunteering activity when opted for the same. This encourages greater volunteer participation and ensures volunteer availability for the demanding routes and trips. It also dampens the fluctuation of the availability of the volunteers over time.

Our work encourages the volunteers to remain active. Otherwise, they do not get their incentive amounts. A competitive participation from the volunteers will keep the surplus distribution movement active as the availability of volunteers keeps steady over time. However, one limitation of our work is that although our work takes into account the number of surplus deliveries of the volunteers, it does not account for the distance travelled by the volunteer while delivering the surplus. It may so happen that, the distance travelled by an intercity volunteer is far more than a volunteer who has more number of local trips during the same time window. This effort needs to be rewarded in a future work so that surplus distribution can truly go across borders. Another limitation of our paper is that, here, incentives to the volunteers are considered in terms of monetary benefits only. However, many volunteers may be motivated if non-monetary incentives (e.g., social badges, etc.) are provided. A probe for a hybrid mechanism, considering both monetary and non-monetary incentives, will be a nice future direction. Information dissemination or information nudges are an important aspect in behavioural economics. In our case, how different chains of volunteers are performing is not known to each other. If we dispense this information to each other in a systematic manner and embed the idea into our algorithm, the volunteers may be more motivated by seeing each other's activities. This is another future scope of our paper.

Funding

Fatos Xhafa's work is supported by Project "Optimizing Carsharing and Ridesharing Mobility in Smart Sustainable Cities" (grant 21S09355-001) of the Ajuntament of Barcelona and the Fundació "la Caixa" under the framework of the Barcelona Science Plan 2020–2023.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We used real data taken from a real life smart city context in OPEN DATA BARCELONA at "<https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us>" and cast it to our settings.

References

- Adenso-Díaz, B., Lozano, S., & Palacio, A. (2017). Effects of dynamic pricing of perishable products on revenue and waste. *Applied Mathematical Modelling*, 45, 148–164.
- Ahad, M. A., Paiva, S., Tripathi, G., & Feroz, N. (2020). Enabling technologies and sustainable smart cities. *Sustainable Cities and Society*, 61, Article 102301.
- Alam, M., Abid, F., Guangpei, C., & Yunrong, L. (2020). Social media sentiment analysis through parallel dilated convolutional neural network for smart city applications. *Computer Communications*, 154, 129–137.
- Alblihed, N., Almutairi, M., Almahmoud, R., Aladhadh, S., & Alabdulatif, A. (2022). Developing food charity operations management system. In *2022 2nd international conference on computing and information technology* (pp. 93–96). IEEE.
- Ananprakit, N., & Esbjörnsson, M. (2017). Traceability in charitable food redistribution system-ensuring food safety and quality in a cold chain. Student Paper.
- Arcuri, S. (2019). Food poverty, food waste and the consensus frame on charitable food redistribution in Italy. *Agriculture and Human Values*, 36(2), 263–275.
- Babaioff, M., Dobzinski, S., Oren, S., & Zohar, A. (2012). On bitcoin and red balloons. In *Proceedings of the 13th ACM conference on electronic commerce* (pp. 56–73).
- Bergström, P., Malefors, C., Strid, I., Hanssen, O. J., & Eriksson, M. (2020). Sustainability assessment of food redistribution initiatives in Sweden. *Resources*, 9(3), 27.
- Berns, K., & Rossitto, C. (2019). From commodities to gifts: Redistributing surplus food locally. In *Ethnographies of collaborative economies, Edinburgh, Scotland, 25 October, 2019*.
- Berry, B., & Acheson, A. (2017). Sharing isn't easy: Food waste and food redistribution in maine K–12 schools. *Maine Policy Review*, 26(1), 47–58.
- Chen, M., Jiang, Y., Guizani, N., Zhou, J., Tao, G., Yin, J., et al. (2020). Living with I-fabric: Smart living powered by intelligent fabric and deep analytics. *IEEE Network*, 34(5), 156–163.
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). *Introduction to algorithms*. MIT Press.
- Dahbi, M., Saadane, R., & Mbarki, S. (2019). Social media sentiment monitoring in smart cities: an application to moroccan dialects. In *Proceedings of the 4th international conference on smart city applications* (pp. 1–6).
- Davies, A. R., Cretella, A., & Franck, V. (2019). Food sharing initiatives and food democracy: Practice and policy in three European cities. *Politics and Governance*, 7(4), 8–20.
- Davies, A. R., Edwards, F., Marovelli, B., Morrow, O., Rut, M., & Weymes, M. (2017). Making visible: Interrogating the performance of food sharing across 100 urban areas. *Geoforum*, 86, 136–149.
- Davies, A., & Evans, D. (2019). Urban food sharing: Emerging geographies of production, consumption and exchange. *Geoforum*, 99, 154–159.
- Davies, A. R., & Legg, R. (2018). Fare sharing: Interrogating the nexus of ICT, urban food sharing, and sustainability. *Food, Culture & Society*, 21(2), 233–254.
- Davies, A., & Weymes, M. (2018). [Re] valuing surplus: Transitions, technologies and tensions in redistributing prepared food in san francisco. *Geoforum*.
- De Guimarães, J. C. F., Severo, E. A., Júnior, L. A. F., Da Costa, W. P. L. B., & Salmoria, F. T. (2020). Governance and quality of life in smart cities: Towards sustainable development goals. *Journal of Cleaner Production*, 253, Article 119926.
- Demirag, O. C., Kumar, S., & Rao, K. M. (2017). A note on inventory policies for products with residual-life-dependent demand. *Applied Mathematical Modelling*, 43, 647–658.
- Duan, Y., Cao, Y., & Huo, J. (2018). Optimal pricing, production, and inventory for deteriorating items under demand uncertainty: The finite horizon case. *Applied Mathematical Modelling*, 58, 331–348.
- Durr, S. J., & Lotz-Sistika, H. (2019). Exploring the social learning value enabled by affordances of the food for us mobile application: The story of a South African food redistribution app. *International Association for Development of the Information Society*.
- El Bilali, H., & Allahyari, M. S. (2018). Transition towards sustainability in agriculture and food systems: Role of information and communication technologies. *Information Processing in Agriculture*, 5(4), 456–464.
- Eriksson, M., Giovannini, S., & Ghosh, R. K. (2020). Is there a need for greater integration and shift in policy to tackle food waste? Insights from a review of European union legislations. *SN Applied Sciences*, 2(8), 1–13.
- Esteso, A., Alemany, M., & Ortiz, Á. (2021). Impact of product perishability on agri-food supply chains design. *Applied Mathematical Modelling*.

- Facchini, E., Iacovidou, E., Gronow, J., & Voulvoulis, N. (2018). Food flows in the United Kingdom: The potential of surplus food redistribution to reduce waste. *Journal of the Air & Waste Management Association*, 68(9), 887–899.
- Falcone, P. M., & Imbert, E. (2017). Bringing a sharing economy approach into the food sector: The potential of food sharing for reducing food waste. In *Food waste reduction and valorisation* (pp. 197–214). Springer.
- Fleming, M. (2017). Estimating environmental emissions from produce waste redistribution in the capital region. Thesis Work.
- Frigo, A., & Lucchini, M. (2018). Working together towards circular economy: recovery and redistribution of surplus food for social purposes. In *6th international conference on sustainable solid waste management which takes place on the Island of Naxos, Greece*. Furbank, L. (2016). 59 Organizations fighting food loss and waste. <https://foodtank.com/news/2016/07/59-organizations-fighting-food-loss-and-waste/>.
- Gahlot, S., Reddy, S., & Kumar, D. (2018). Review of smart health monitoring approaches with survey analysis and proposed framework. *IEEE Internet of Things Journal*, 6(2), 2116–2127.
- Ganglbauer, E., Fitzpatrick, G., & Molzer, G. (2012). Creating visibility: understanding the design space for food waste. In *Proceedings of the 11th international conference on mobile and ubiquitous multimedia* (pp. 1–10).
- Geraskin, M. (2021). Choice of optimal incentive system for large social groups of volunteers. In *2021 international conference on information technology and nanotechnology* (pp. 1–7). IEEE.
- Han, M. J. N., & Kim, M. J. (2021). A critical review of the smart city in relation to citizen adoption towards sustainable smart living. *Habitat International*, 108, Article 102312.
- Harvey, J., Smith, A., Goulding, J., & Illodo, I. B. (2020). Food sharing, redistribution, and waste reduction via mobile applications: A social network analysis. *Industrial Marketing Management*, 88, 437–448.
- Hassankhani, M., Alidadi, M., Sharifi, A., & Azhdari, A. (2021). Smart city and crisis management: Lessons for the COVID-19 pandemic. *International Journal of Environmental Research and Public Health*, 18(15), 7736.
- Herbon, A. (2017). A non-cooperative game model for managing a multiple-aged expiring inventory under consumers' heterogeneity to price and time. *Applied Mathematical Modelling*, 51, 38–57.
- Hodorog, A., Petri, I., & Rezgui, Y. (2022). Machine learning and natural language processing of social media data for event detection in smart cities. *Sustainable Cities and Society*, 85, Article 104026.
- Ismagilova, E., Hughes, L., Dwivedi, Y. K., & Raman, K. R. (2019). Smart cities: Advances in research—An information systems perspective. *International Journal of Information Management*, 47, 88–100.
- Kamruzzaman, M. (2020). Architecture of smart health care system using artificial intelligence. In *2020 IEEE international conference on multimedia & expo workshops* (pp. 1–6). IEEE.
- King, E. (2019). Identifying arkansas food desert blocks suitable for a peer-to-peer modeled food redistribution program. Thesis Work.
- Koivunen, L., Laato, S., Rauti, S., Naskali, J., Nissilä, P., Ojansivu, P., et al. (2020). Increasing customer awareness on food waste at university cafeteria with a sensor-based intelligent self-serve lunch line. In *2020 IEEE international conference on engineering, technology and innovation (ICE/ITMC)* (pp. 1–9). IEEE.
- Kor, A.-L., Yanovsky, M., Pattinson, C., & Kharchenko, V. (2016). SMART-ITEM: Iot-enabled smart living. In *2016 future technologies conference* (pp. 739–749). IEEE.
- Kumar, A., Krishnamurthi, R., Nayyar, A., Sharma, K., Grover, V., & Hossain, E. (2020). A novel smart healthcare design, simulation, and implementation using healthcare 4.0 processes. *IEEE Access*, 8, 118433–118471.
- Kummita, R. K. R., & Crutzen, N. (2017). How do we understand smart cities? An evolutionary perspective. *Cities*, 67, 43–52.
- Kuzior, A., Krawczyk, D., Brożek, P., Pakhnenko, O., Vasylieva, T., & Lyeonov, S. (2022). Resilience of smart cities to the consequences of the COVID-19 pandemic in the context of sustainable development. *Sustainability*, 14(19), 12645.
- Lai, C. S., Jia, Y., Dong, Z., Wang, D., Tao, Y., Lai, Q. H., et al. (2020). A review of technical standards for smart cities. *Clean Technologies*, 2(3), 290–310.
- Leelaarporn, P., Wachiraphan, P., Kaewlee, T., Udsa, T., Chaisaen, R., Chok-satchawathi, T., et al. (2021). Sensor-driven achieving of smart living: A review. *IEEE Sensors Journal*, 21(9), 10369–10391.
- Light, A., & Miskelly, C. (2019). Platforms, scales and networks: meshing a local sustainable sharing economy. *Computer Supported Cooperative Work (CSCW)*, 28(3–4), 591–626.
- Lopes, N. V. (2017). Smart governance: A key factor for smart cities implementation. In *2017 IEEE international conference on smart grid and smart cities* (pp. 277–282). IEEE.
- Mousa, T. Y., & Freeland-Graves, J. H. (2017). Organizations of food redistribution and rescue. *Public Health*, 152, 117–122.
- Nagarajan, S. M., Deverajan, G. G., Chatterjee, P., Alnumay, W., & Muthukumaran, V. (2022). Integration of IoT based routing process for food supply chain management in sustainable smart cities. *Sustainable Cities and Society*, 76, Article 103448.
- Nikravech, M., Kwan, V., Dobernig, K., Wilhelm-Rechmann, A., & Langen, N. (2020). Limiting food waste via grassroots initiatives as a potential for climate change mitigation: a systematic review. *Environmental Research Letters*.
- Philip, D., Hod-Ovadia, S., & Troen, A. M. (2017). A technical and policy case study of large-scale rescue and redistribution of perishable foods by the “Leket Israel” food bank. *Food and Nutrition Bulletin*, 38(2), 226–239.
- Prova, R. R., Rayhan, A., Shilon, R. S., & Khan, M. M. (2021). A web and mobile based approach to redistribute consumable food waste. In *2021 12th international conference on computing communication and networking technologies* (pp. 1–6). IEEE.
- Puri, M., Varde, A., Du, X., & De Melo, G. (2018). Smart governance through opinion mining of public reactions on ordinances. In *2018 IEEE 30th international conference on tools with artificial intelligence* (pp. 838–845). IEEE.
- Rombach, M. (2018). New perspectives on food redistribution: Evidence from the German Food Bank and other volunteer based organizations (Ph.D. thesis), Technische Universität München.
- Rut, M., & Davies, A. R. (2018). Sharing foodscapes: shaping urban foodscapes through messy processes of food sharing. In *Messy ethnographies in action* (pp. 167–175). Vernon Press Wilmington, DE.
- Rut, M., Davies, A. R., & Ng, H. (2020). Participating in food waste transitions: exploring surplus food redistribution in Singapore through the ecologies of participation framework. *Journal of Environmental Policy & Planning*, 1–14.
- Sanyal, S., Kumar Singh, V., Xhafa, F., Sanyal, B., & Mukhopadhyay, S. (2021). A game theoretic framework for surplus food distribution in smart cities and beyond. *Applied Sciences*, 11(11), 5058.
- Sanyal, S., Mukhopadhyay, S., Xhafa, F., Kumar Singh, V., & Sanyal, B. (2022). An incentive framework for surplus food distribution in the context of smart cities (accepted). In *The 25th international conference on network-based information systems (NBIS-2022)*. Springer.
- Shi, Z. R., Yuan, Y., Lo, K., Lizarondo, L., & Fang, F. (2020). Improving efficiency of volunteer-based food rescue operations. In *AAAI* (pp. 13369–13375).
- Spicer, L. C., Patel, D., Bodurtha, E. R., & Maslen, A. M. (2020). Surplus food management: A case study. Student Work.
- Spring, C., Adams, M., & Hardman, M. (2019). Sites of learning: Exploring political ecologies and visceral pedagogies of surplus food redistribution in the UK. *Policy Futures in Education*, 17(7), 844–861.
- Spring, C. A., & Biddulph, R. (2020). Capturing waste or capturing innovation? Comparing self-organising potentials of surplus food redistribution initiatives to prevent food waste. *Sustainability*, 12(10), 4252.
- Umair, M., Cheema, M. A., Cheema, O., Li, H., & Lu, H. (2021). Impact of COVID-19 on IoT adoption in healthcare, smart homes, smart buildings, smart cities, transportation and industrial IoT. *Sensors*, 21(11), 3838.
- Vittuari, M., De Menna, F., Gaiani, S., Falasconi, L., Politano, A., Dietershagen, J., et al. (2017). The second life of food: An assessment of the social impact of food redistribution activities in emilia romagna, Italy. *Sustainability*, 9(10), 1817.
- Weymes, M., & Davies, A. R. (2018). Disruptive technologies? Scaling relational geographies of ICT-mediated surplus food redistribution. *Tech. rep.*, SHARECITY working paper.
- Weymes, M., & Davies, A. R. (2019). [Re] valuing surplus: Transitions, technologies and tensions in redistributing prepared food in san francisco. *Geoforum*, 99, 160–169.
- Zhong, R., Xu, X., & Wang, L. (2017). Food supply chain management: systems, implementations, and future research. *Industrial Management & Data Systems*.